Sec 9.6 Functions of two variables.



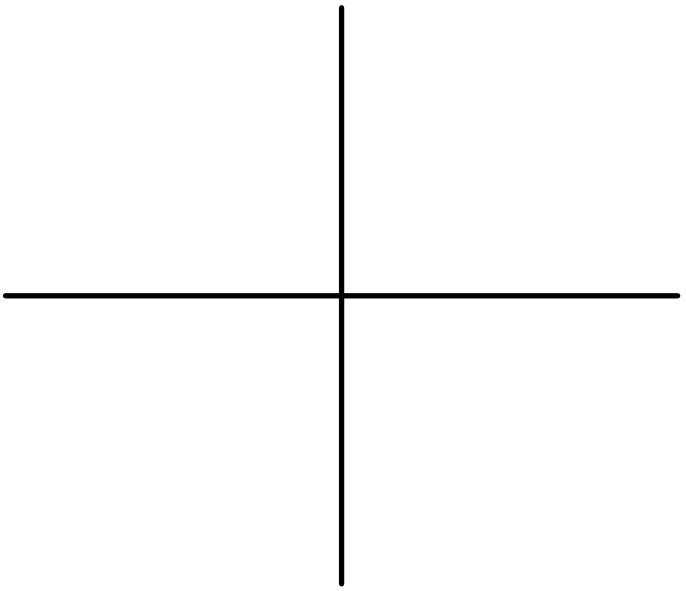
Two variables … HA.. HA..HA...

Definition A **function** f **of two variables** is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the **domain** of f and its **range** is the set of values that f takes on, that is, $\{f(x, y) \mid (x, y) \in D\}$.

Domain: The domain of a function of two variables is the set of all ordered pairs (x,y) such that(x, y, f(x,y)) can be plotted in R^3 and f(x,y) is calculable.

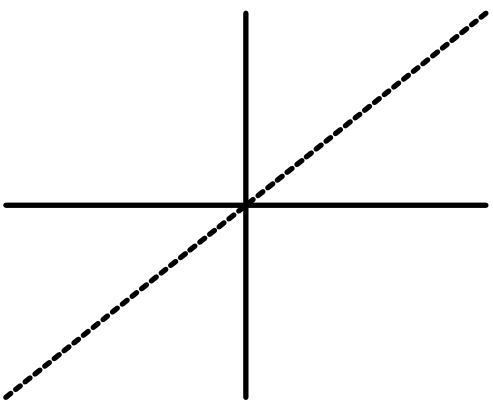
Ex. Find and sketch the domain of

$$f(x,y) = \frac{\sqrt{x-y}}{y-2}$$



Range: The range of a function of two variables are all of the values z=f(x,y) which come from all of the ordered pairs (x,y) in the Domain of f.

Ex Find the range of $f(x,y) = x^2 + y^2$



Sec 9.6 Functions of two variables.

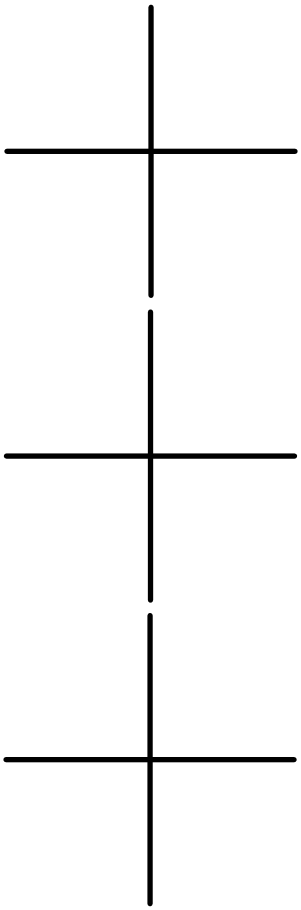
How do we graph functions of two variables? We can do this using Level Sets or Traces as Stewart calls them.

A Level Set (Trace): For a function of two variables we may fix either an x , y , or z vale in our equation which puts us in a specific plane. We can then study the intersection of the plane and our surface in that plane. This intersection is the level set or trace..

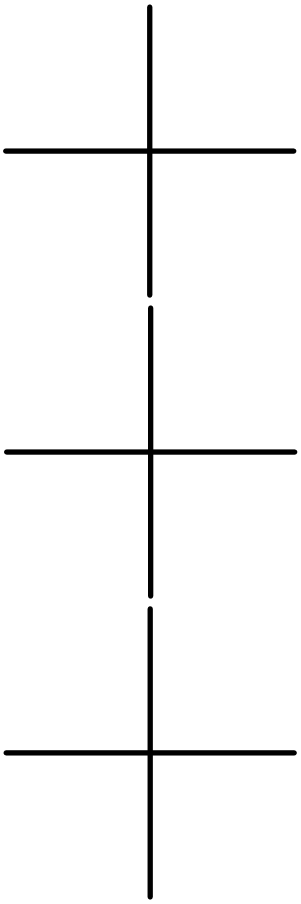
Example: Consider the two variable function $f(x,y) = x^2 + y^2$

Calculate and graph Level sets for z=0 , 1 , 4 Do the same for y=0 , 1 , 4 and x=0,1,4 note z=f(x,y)

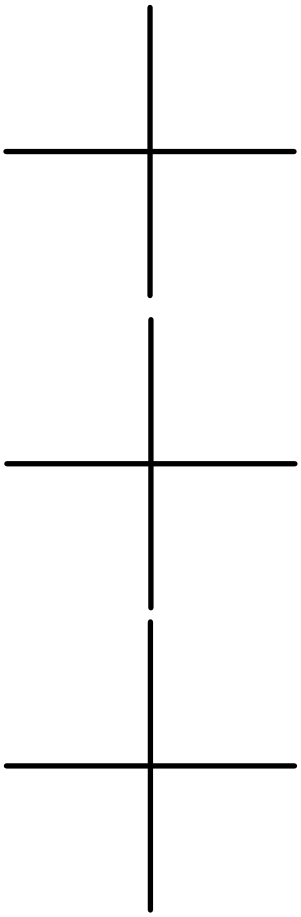
Fixing Z values

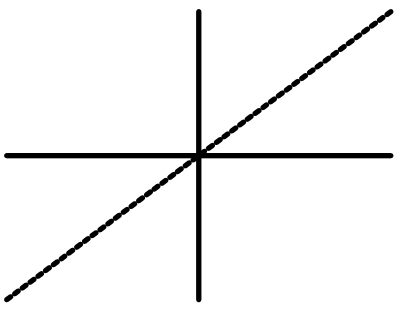


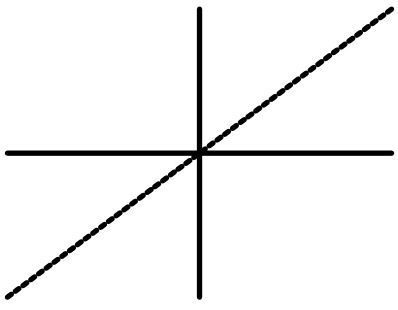
Fixing y values

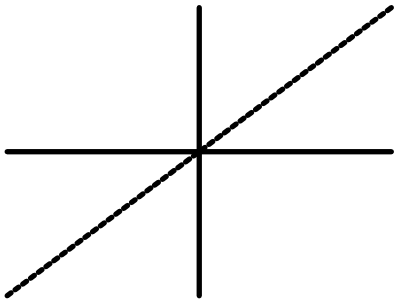


Fixing x values





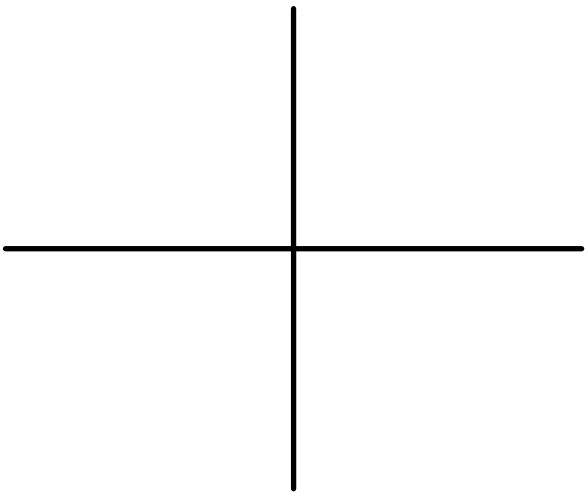




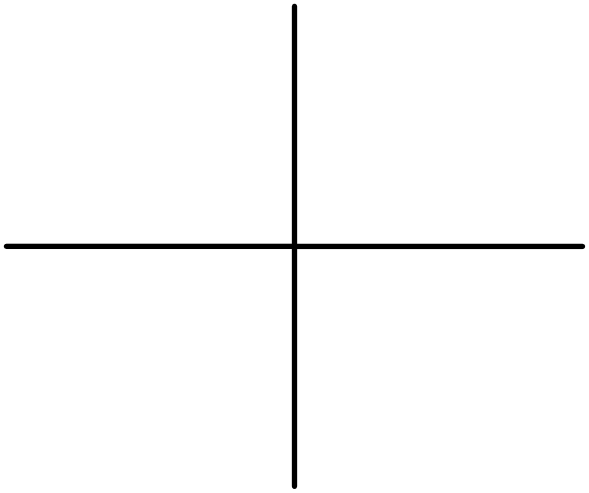
Sec 9.6 Functions of two variables.

Example Use Traces to sketch the graph of the following function on the set
{(x,y)|0<x<2pi , 0<y<2pi}

$f(x,y) = cos(x) + sin(y) \quad Try \;\; x = 0, \pi, 2\pi \;\; and \; y = 0, \pi, 2\pi$

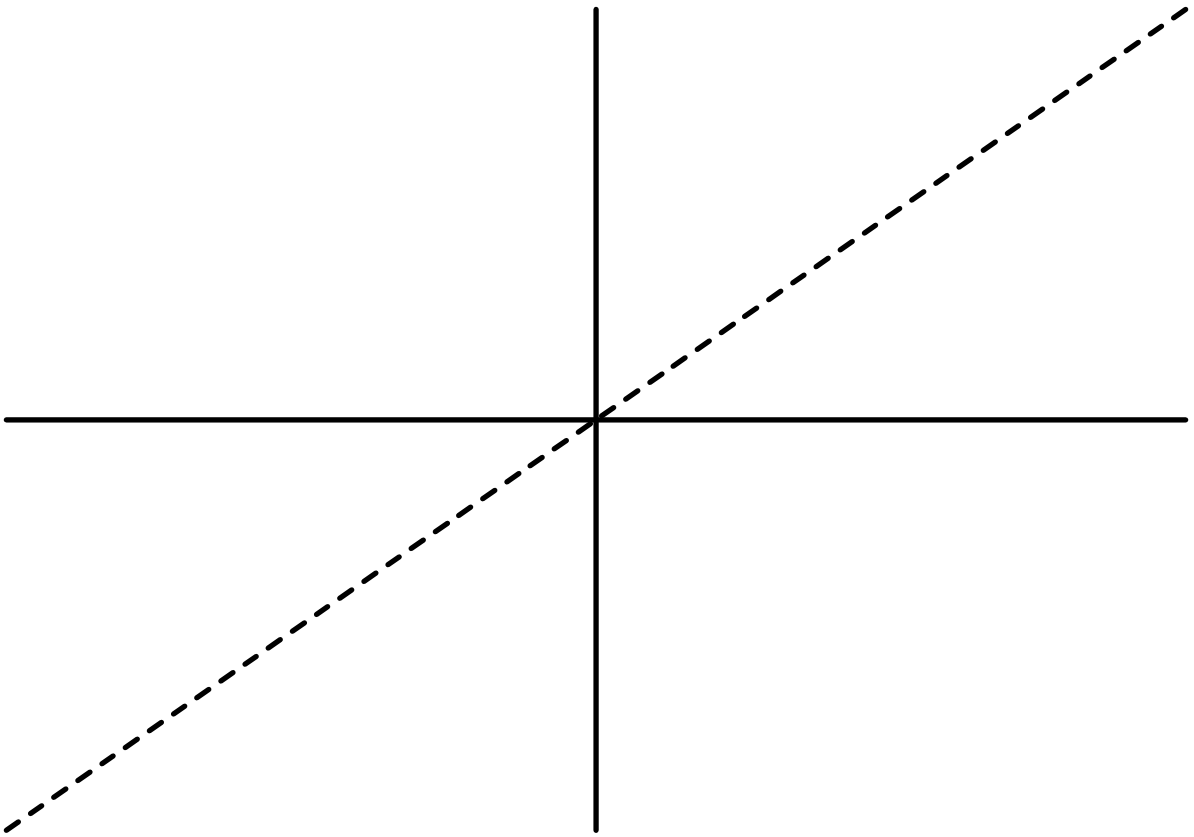


X-level sets



Y level sets

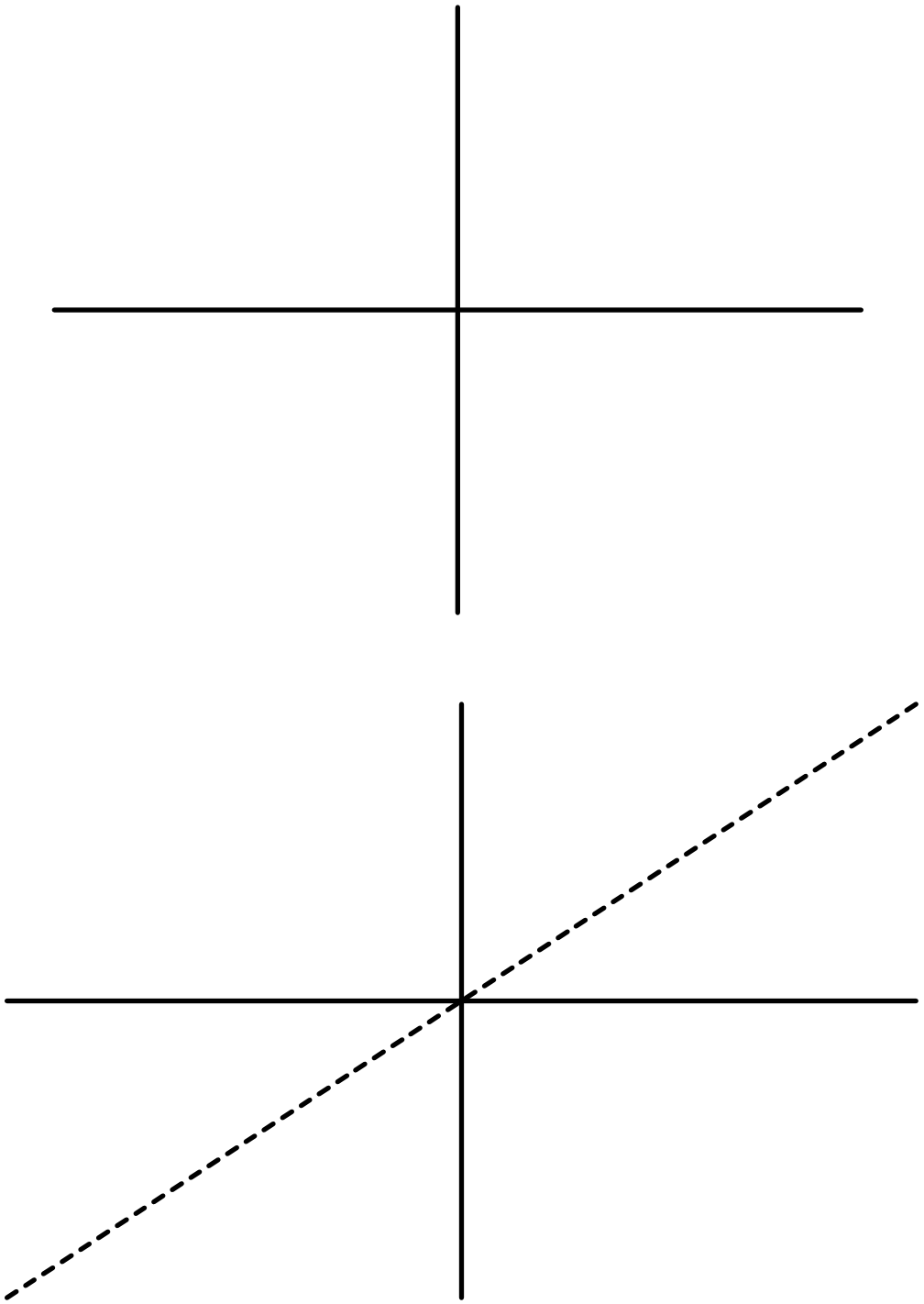
Put it all together...



http://www.math.uri.edu/~bkaskosz/flashmo/graph3d/

Sec 9.6 Functions of two variables.

The difference between $f(x) = x^2 \quad and \quad f(x,y) = y^2$

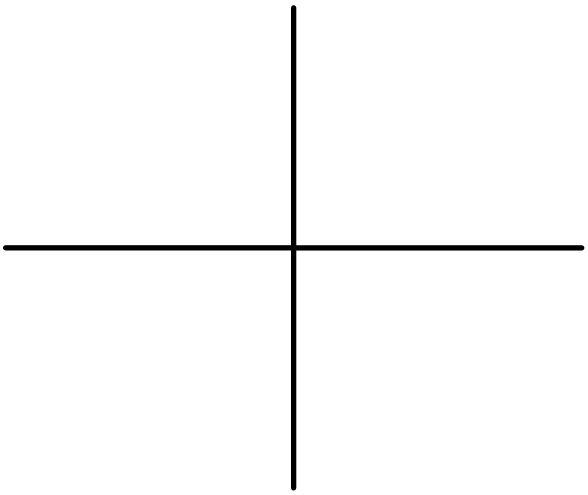


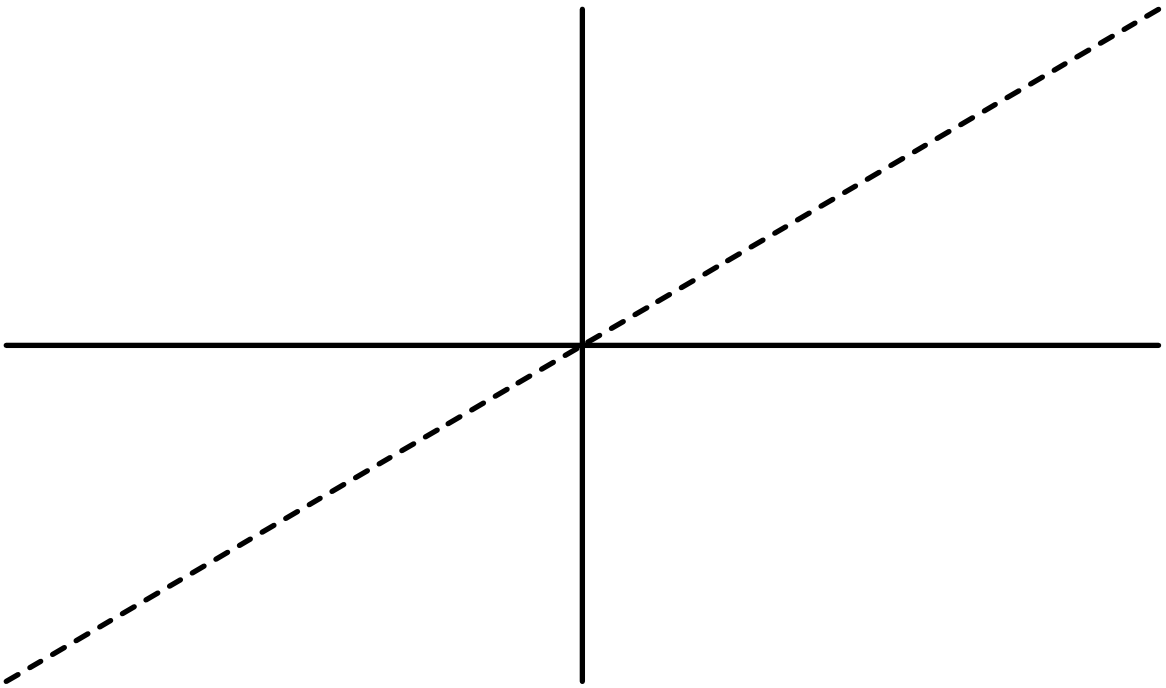
Since the graph of $f(x,y) = x^2$ Consists of points (x,y,z) and there is no dependence on y, the y values may be what ever we like and it will not effect the z values…. In this way the parabola is just stretched in the y-direction for ever. This is called a parabolic cylinder

Sec 9.6 Functions of two variables.

HW example Sketch the graph.

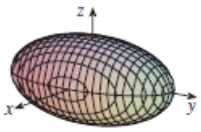
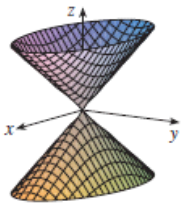
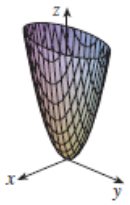
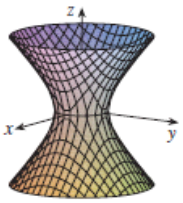
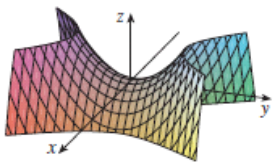
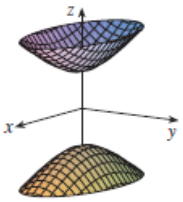
13. $f(x, y) = y^2 + 1$





Sec 9.6 Functions of two variables.

Quadratic Surfaces...

Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ All traces are ellipses. If $a=b=c$, the ellipsoid is a sphere.	Cone 	$\frac{z^2}{c^2}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x=k$ and $y=k$ are hyperbolas if $k\neq 0$ but are pairs of lines if $k=0$.
Elliptic Paraboloid 	$\frac{z}{c}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	Hyperboloid of One Sheet 	$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c}=\frac{x^2}{a^2}-\frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c<0$ is illustrated.	Hyperboloid of Two Sheets 	$-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ Horizontal traces in $z=k$ are ellipses if $k>c$ or $k<-c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.

HW Example Which one is it?

21. $4x^2 + y^2 + 4z^2 - 4y - 24z + 36 = 0$

Sec 9.6 Functions of two variables.

Graphing a plane…..

Sketch a graph of the plane $2x+3y+6z=4$